INTELLIGENT TRANSPORT SYSTEMS (ITS-401)
COURSE LECTURE NOTES & SYLLABUS REFERENCE
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

UNIT 1: INTRODUCTION TO INTELLIGENT TRANSPORT SYSTEMS

1.1 Fundamentals and Architecture of ITS

Intelligent Transport Systems (ITS) integrate advanced information, communication, electronics, and sensor technologies into transportation networks to enhance mobility, improve road safety, and minimize environmental impact. The national ITS architecture consists of physical, logical, and communication layers that interconnect traffic management centers, roadside field equipment, vehicles, and commuters.

1.2 Sensor Technologies and Traffic Surveillance

Traffic data collection utilizes both intrusive and non-intrusive sensors. Inductive loop detectors embedded beneath pavement detect magnetic field disruptions caused by metallic vehicle chassis, providing reliable vehicle counts, speed, and occupancy metrics. Non-intrusive sensors include pneumatic road tubes, microwave radars, infrared sensors, and computer vision closed-circuit television (CCTV) cameras. Video Image Processing Systems (VIPS) employ deep learning for automated incident detection and license plate recognition.

UNIT 2: ADVANCED TRAFFIC MANAGEMENT SYSTEMS (ATMS)

2.1 Real-Time Traffic Signal Optimization

Advanced Traffic Management Systems (ATMS) dynamically adjust traffic signal timings based on real-time saturation levels. SCATS (Sydney Coordinated Adaptive Traffic System) and SCOOT (Split Cycle Offset Optimisation Technique) continually calculate green splits, cycle lengths, and coordinate intersection offsets to minimize vehicular delay and stop frequencies across arterial corridors.

2.2 Incident Detection and Ramp Metering

Automated incident detection algorithms monitor flow fluctuations. Freeway ramp metering utilizes traffic signals on entrance ramps to regulate the rate at which vehicles enter mainline highways, preventing freeway breakdown and shockwave propagation during peak demand periods. Variable Message Signs (VMS) disseminate real-time advisory alerts and lane closure warnings.

UNIT 3: ADVANCED TRAVELER INFORMATION SYSTEMS (ATIS)

3.1 Architecture of ATIS and Commuter Guidance

Advanced Traveler Information Systems (ATIS) deliver pre-trip and en-route travel information to road users. By processing historical traffic baselines alongside live crowdsourced probe GPS feeds, ATIS platforms provide dynamic route guidance, estimated times of arrival (ETA), and multimodal transit alternatives.

3.2 Electronic Toll Collection (ETC) and RFID

Automated Electronic Toll Collection systems, such as FASTag, leverage Radio Frequency Identification (RFID) operating in the passive UHF band (865-868 MHz). As an RFID-tagged vehicle crosses an electronic toll plaza at speed, overhead transceivers read tag IDs and automatically debit linked prepaid bank wallets, eliminating manual cash delays and reducing emissions.

UNIT 4: CONNECTED AND AUTONOMOUS VEHICLES (CAV)

4.1 V2X Communication Frameworks

Vehicle-to-Everything (V2X) communication encompasses Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Vehicle-to-Pedestrian (V2P) exchanges. Dedicated Short-Range Communications (DSRC) is based on the IEEE 802.11p standard operating in the licensed 5.9 GHz ITS band, offering ultra-low latency (<5 ms) over short ranges (~300 meters). Cellular-V2X (C-V2X) utilizes 4G LTE and 5G sidelink PC5 direct communication, providing longer range and greater bandwidth for sensor sharing.

4.2 Cooperative Adaptive Cruise Control and Platooning

Cooperative Adaptive Cruise Control (CACC) combines radar sensing with real-time V2V telemetry (acceleration, braking commands) to permit heavy vehicles to travel in tightly coupled platoons with sub-second following gaps, dramatically reducing aerodynamic drag and highway congestion.